

IMO Training – Number Theory

Christmas Problem Set

Longtin Chan
longtinclt@gmail.com

28 December, 2019

Instructions:

- Enjoy Maths!
- Merry Christmas and Happy New Year!

1. (Germany 2018)

Define a sequence of integers (a_n) by $a_1 = 1$, and $\forall k > 1, a_k$ is the largest prime factor of $a_1 a_2 \cdots a_{k-1} + 1$. Show that 11 is not a member of the sequence.

2. (Canada 2019)

Let $a, b \in \mathbb{N}$ s.t. $a^2 + 3ab + 3b^2 - 1 \mid a + b^3$. Prove that $a^2 + 3ab + 3b^2 - 1$ is divisible by the cube of an integer greater than 1.

3. (Iran 2019)

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ so that for any distinct $x, y, z \in \mathbb{N}$, $x + y + z$ is a perfect square if and only if $f(x) + f(y) + f(z)$ is a perfect square.

4. (Korea 2019)

Assume that $m, n, k \in \mathbb{N}$ satisfy the equations $m^2 + 1 = 2n^2$ and $2m^2 + 1 = 11k^2$. Find the remainder when n is divided by 17.

5. (Greece 2019)

Find all $x, y \in \mathbb{Q}^+$ s.t. $yx^y = y + 1$.